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**Project Title:** Script-Controlled ACL – Restrict Record Access Based on Field Value

**Date:** 25 September 2026

**Team ID:** SWTID-2026-2214

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## **PHASE 7 – PROJECT DOCUMENTATION**

### **1. Introduction:**

This project implements a secure record-access mechanism using **Access Control Lists (ACLs)** in ServiceNow.

The system controls access to the **Institution Details** table based on user roles and branch information. Different roles are used for Read, Create, Write, and Delete operations.

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### **2. Problem Statement:**

Organizations store sensitive student and institutional information in databases. Without proper access control, unauthorized users may view, create, modify, or delete records.

The project addresses this problem by implementing role-based and branch-based access restrictions in ServiceNow.

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### **3. Project Objectives:**

The main objectives are:

- To implement ACLs in ServiceNow.
- To restrict unauthorized record access.

- To implement role-based permissions.
  - To restrict Read access based on the Branch field.
  - To control Create, Write, and Delete operations.
  - To use scripting for customized access control.
  - To provide administrators with full access.
  - To improve the security of institutional information.
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#### 4. Technologies and Tools:

| Tool         | Purpose                           |
|--------------|-----------------------------------|
| ServiceNow   | Project implementation            |
| JavaScript   | ACL scripting                     |
| Web Browser  | Accessing ServiceNow              |
| GitHub       | Project repository and submission |
| Google Drive | Demonstration video               |

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#### 5. System Design:

The system follows:

**User → Role → ACL → Script/Data Condition → Access Decision**

For Read access, the system checks the user's role and applies the branch condition.

The main Read condition is:

**Branch is EEE**

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## 6. Database/Table Design:

### Institution Details

**Table Name:** u\_institution\_details

| Field               | Type             |
|---------------------|------------------|
| Student Roll Number | Auto Number      |
| Student Name        | Reference – User |
| Faculty Name        | Reference – User |
| Branch              | Choice           |
| Email               | String           |
| Phone Number        | String           |
| Description         | Multi String     |

### Branch Values

- EEE
- ECE
- CSE

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## 7. Role Design:

The project uses the following roles:

| Role  | Permission  |
|-------|-------------|
| bb1   | Read        |
| bb2   | Create      |
| bb3   | Write       |
| bb4   | Delete      |
| admin | Full access |

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## 8. ACL Implementation:

### Read ACL

**Table:** u\_institution\_details

**Operation:** Read

**Role:** bb1

**Advanced:** Yes

**Data Condition:** Branch is EEE

**Script:**

```
(function () {  
    if (gs.hasRole('admin')) {  
        return true;  
    }  
    if (gs.hasRole('bb1')) {  
        return true;  
    }  
    return false;  
})();
```

The script validates the user's role. The **Branch is EEE** data condition provides the branch-based record restriction.

### Create ACL

**Operation:** Create

**Role:** bb2

### Write ACL

**Operation:** Write

**Role:** bb3

### Delete ACL

**Operation:** Delete

**Role:** bb4

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## 9. System Workflow:

The implemented workflow is:

1. User logs in to ServiceNow.
  2. User accesses the Institution Details table.
  3. ServiceNow checks the user's role.
  4. Applicable ACL is evaluated.
  5. For Read access, the Branch condition is checked.
  6. If the required conditions are satisfied, access is granted.
  7. Otherwise, access is denied.
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## 10. Testing:

Testing was performed using different users, roles, and branch records.

| Test Area                  | Expected Behavior     |
|----------------------------|-----------------------|
| Administrator              | Full access           |
| bb1 user + EEE record      | Read allowed          |
| bb1 user + ECE record      | Restricted            |
| bb1 user + CSE record      | Restricted            |
| User without required role | Access restricted     |
| bb2 user                   | Create permission     |
| bb3 user                   | Write/Edit permission |
| bb4 user                   | Delete permission     |

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## 11. Project Outcome:

The project demonstrates how ServiceNow ACLs can be used to control access to records.

The implementation provides:

- Role-based access control.
  - Branch-based Read restriction.
  - Separate permissions for CRUD operations.
  - Script-controlled authorization.
  - Administrator full access.
  - Protection against unauthorized record access.
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## 12. Benefits

The system provides the following benefits:

1. **Improved Security** – Restricts unauthorized record access.
  2. **Role-Based Control** – Users receive permissions based on their roles.
  3. **Data Protection** – Sensitive records are protected.
  4. **Controlled Operations** – Create, Write, and Delete operations are separately controlled.
  5. **Flexible Configuration** – ACL rules can be modified as requirements change.
  6. **Better Administration** – Administrators can retain complete access.
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## 13. Limitations

- The current Read restriction is designed specifically for the **EEE branch**.
- Additional branch-specific rules would require additional ACL configuration.
- Users must be assigned the appropriate roles to perform their permitted operations.
- Incorrect ACL configuration could affect record accessibility.

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## 14. Future Enhancements:

The project can be enhanced by:

- Adding separate access rules for ECE and CSE.
- Creating a dashboard to monitor access activities.
- Adding more detailed audit and logging mechanisms.
- Creating automated ACL testing.
- Implementing more granular user and department-based restrictions.
- Extending the solution to additional ServiceNow tables.

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## 15. Conclusion:

The **Script-Controlled ACL – Restrict Record Access Based on Field Value** project demonstrates the implementation of secure record-level access control in ServiceNow.

By combining **roles, ACLs, JavaScript, and a Branch data condition**, the system can control who can access and modify institutional records.

The project provides a practical example of how ServiceNow security mechanisms can be customized to meet organizational access-control requirements.

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